

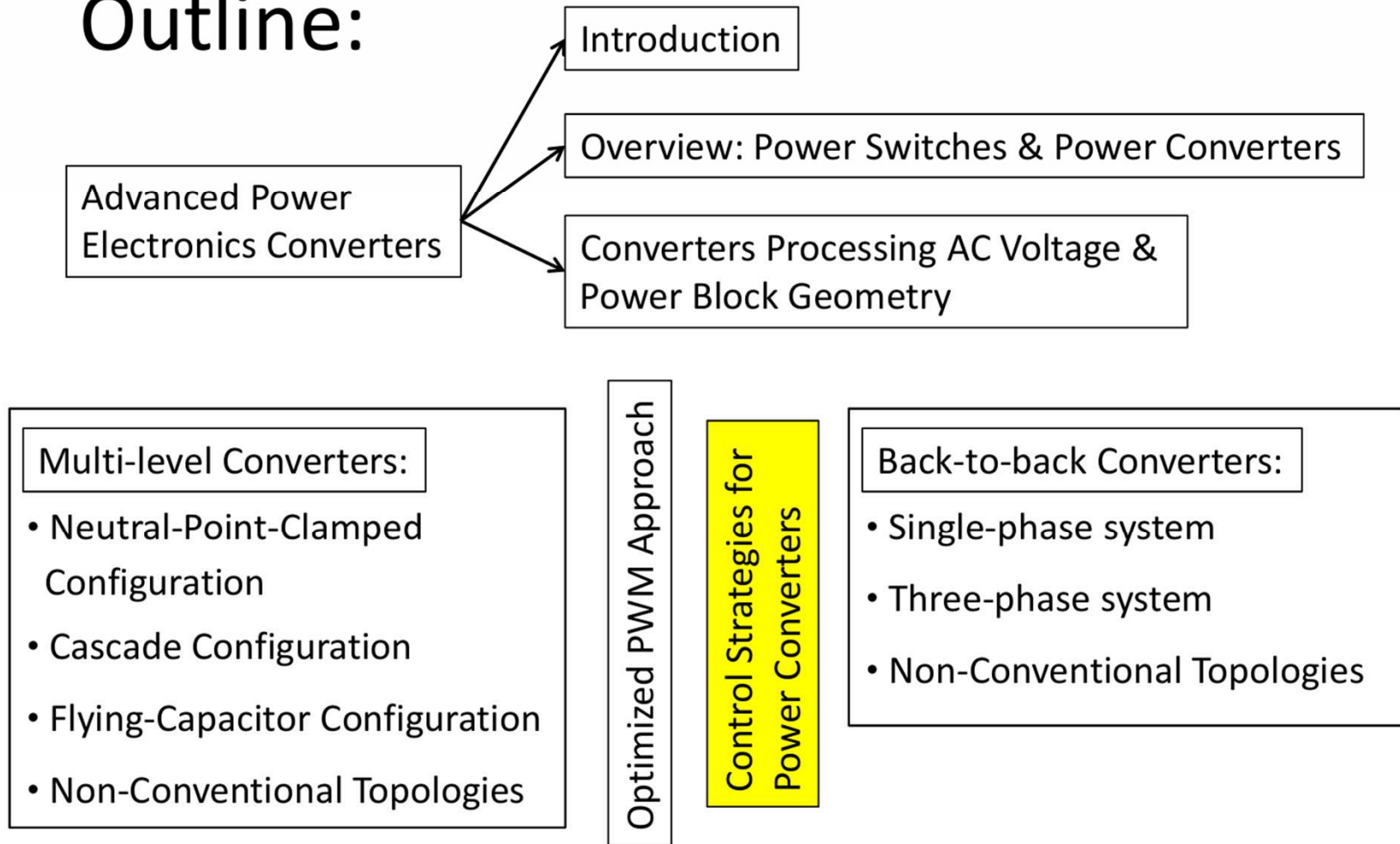
ECE595: Power Converters for AC Systems

Lecture 22 *(Control for Power Converters)*

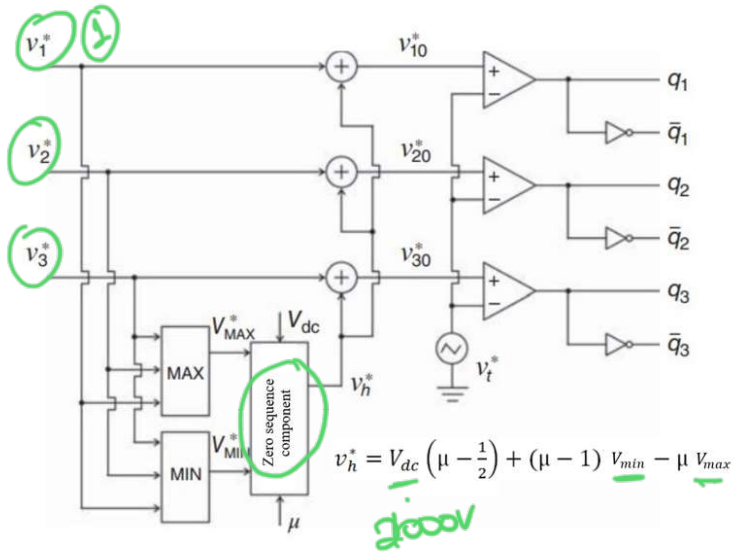
Prof. Euzeli Dos Santos



Outline:

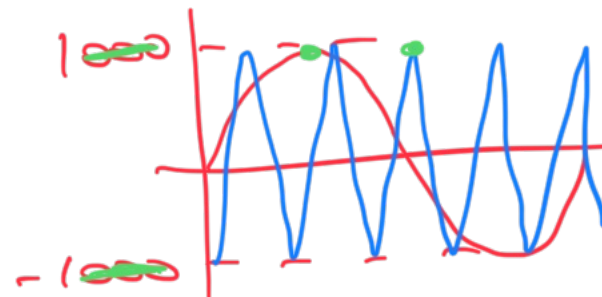
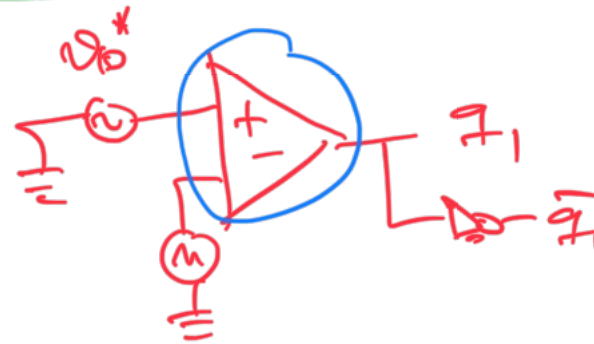
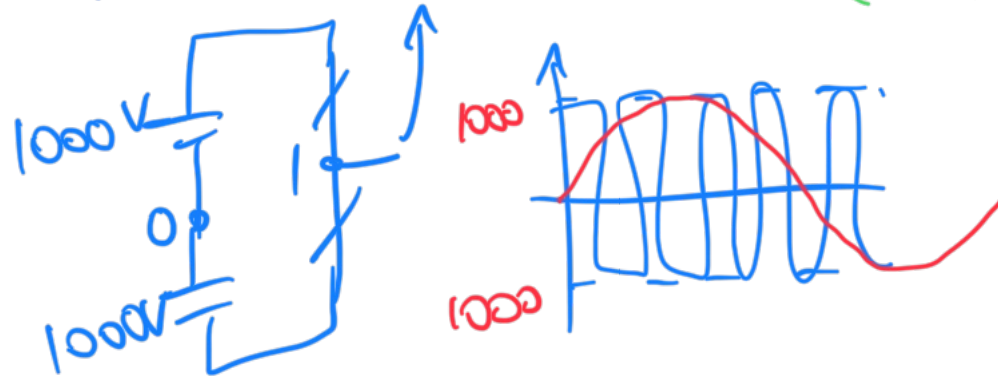


HW 8



pwm: $\langle v_{10} \rangle = \frac{v_{10}^*}{T}$

$\frac{\langle v_{10} \rangle}{(V_{dc}/2)} = v_{10}^*$



```
%begin of the simulation
```

```
while t<tmax
```

```
vs1_ref = V*cos(ws*t);
```

```
vs2_ref = V*cos(ws*t - 2*pi/3);
```

```
vs3_ref = V*cos(ws*t + 2*pi/3);
```

```
%PWM
```

```
if tch < t
```

```
tch = tch + hpwm;
```

```
cont = 0;
```

```
vsdq_ref = inv(Ps)*[vs1_ref;vs2_ref;vs3_ref];
```

```
vsd_ref = vsdq_ref(2);
```

```
vsq_ref = vsdq_ref(3);
```

```
%The six following if's below calculate the times for active vectors for each sector
```

```
if(vs1_ref>=vs2_ref && vs2_ref>=vs3_ref)
```

```
vsdk = sqrt(2/3)*E*cos(0);
```

```
vsqk = sqrt(2/3)*E*sin(0);
```

```
vmdl = sqrt(2/3)*E*cos(pi/3);
```

```
vsq1 = sqrt(2/3)*E*sin(pi/3);
```

```
setor = 1;
```

$k=1$ $h=2$

```
tk = ((vsq1*vsd_ref - vmdl*vsq_ref)*hpwm)/(vsdk*vsq1-vmdl*vsqk);
```

```
t1 = ((vsdk*vsq_ref - vsqk*vsd_ref)*hpwm)/(vsdk*vsq1-vmdl*vsqk);
```

```
to = hpwm - tk - t1;
```

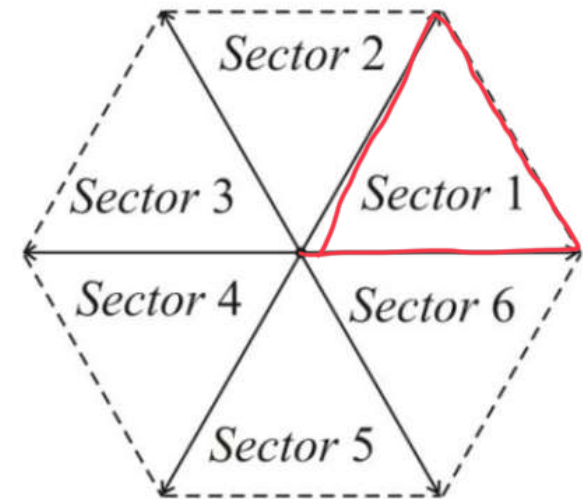
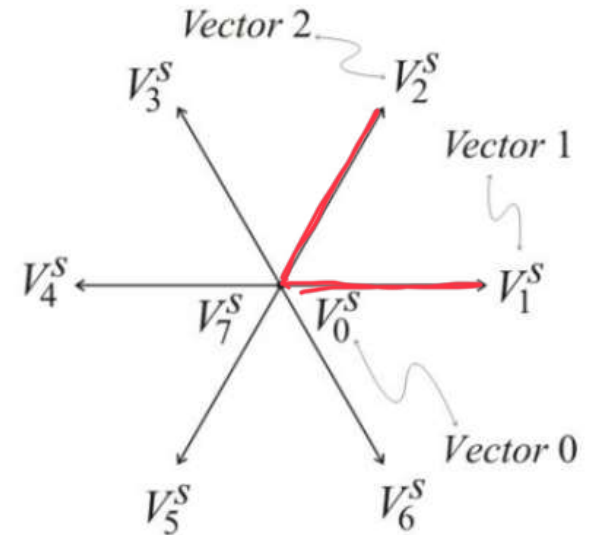
```
toi = m*to;
```

```
tof = (1-m)*to;
```

```
end
```

→ SPWM *
CPWM
→ SVPWM *

15%



$$t_o = t_{oi} + t_{of}$$

μ



%begin of the simulation

while t<tmax

vs1_ref = V*cos(ws*t);

vs2_ref = V*cos(ws*t - 2*pi/3);

vs3_ref = V*cos(ws*t + 2*pi/3);

%PWM

if tch < t

tch = tch + hpwm;

cont = 0;

vsdq_ref = inv(Ps)*[vs1_ref;vs2_ref;vs3_ref];

vsd_ref = vsdq_ref(2);

vsq_ref = vsdq_ref(3);

%The six following if's below calculate the times for active vectors for each sector

if(vs1_ref>=vs2_ref && vs2_ref>=vs3_ref)

vsdk = sqrt(2/3)*E*cos(0);

vsqk = sqrt(2/3)*E*sin(0);

vsdl = sqrt(2/3)*E*cos(pi/3);

vsq1 = sqrt(2/3)*E*sin(pi/3);

setor = 1;

tk = ((vsq1*vsd_ref - vsdl*vsq_ref)*hpwm)/(vsdk*vsq1-vsd1*vsqk);

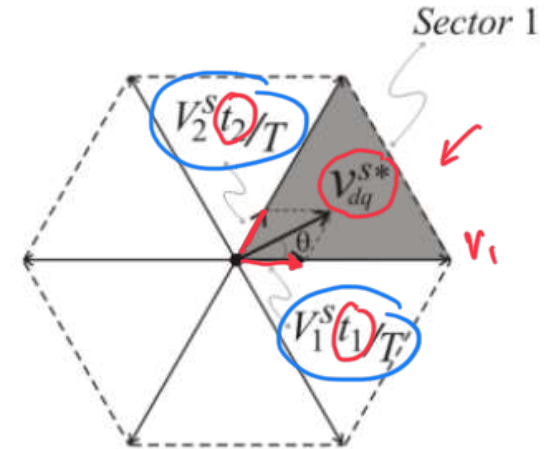
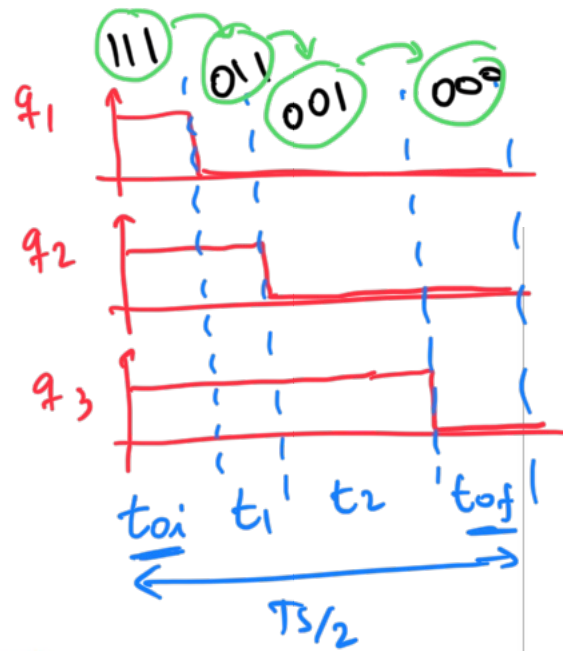
t1 = ((vsdk*vsq_ref - vsqk*vsd_ref)*hpwm)/(vsdk*vsq1-vsd1*vsqk);

to = hpwm - tk - t1;

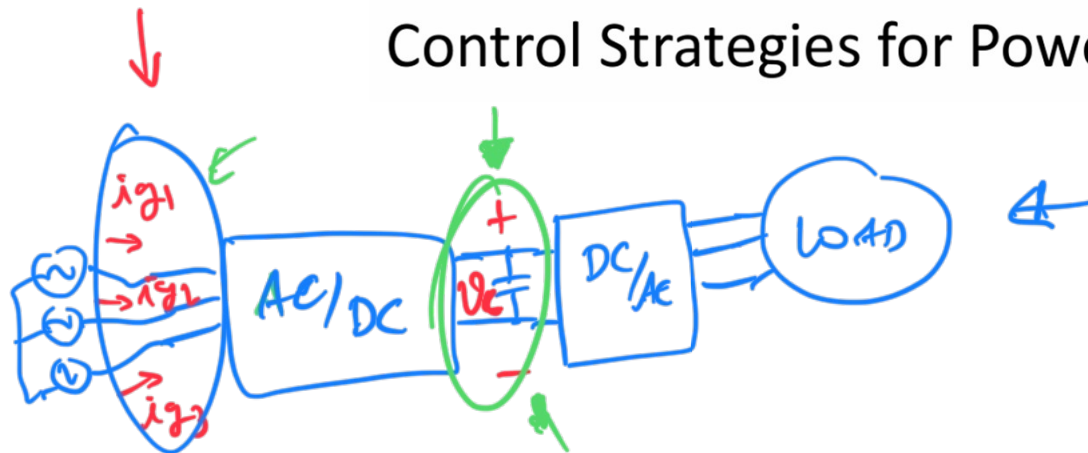
toi = m*to;

tof = (1-m)*to;

end



Control Strategies for Power Converters



THD $\leq 5\%$
PF ≈ 1

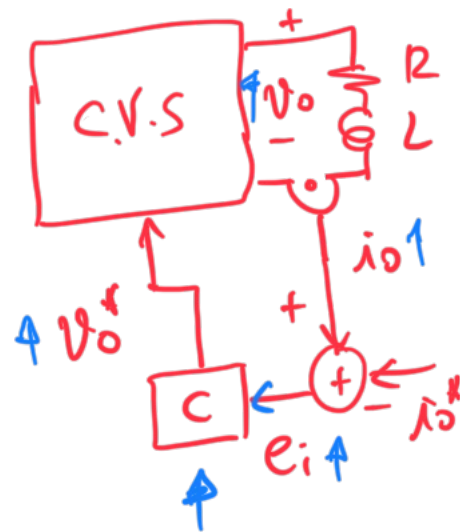
Grid-tie Inv.



$$T_c = f(i_{abc})$$

- CURRENT CONTROL ✓
- DC - LINK voltage control

Control Strategies for Power Converters

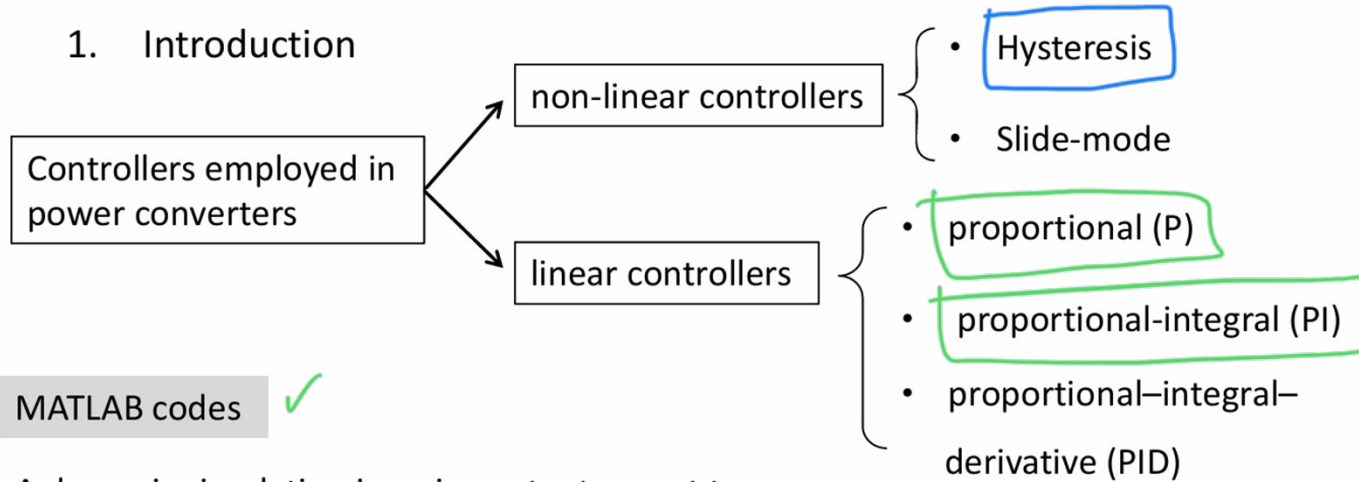


$$e_i = i_o^* - i_o$$



Control Strategies for Power Converters

1. Introduction



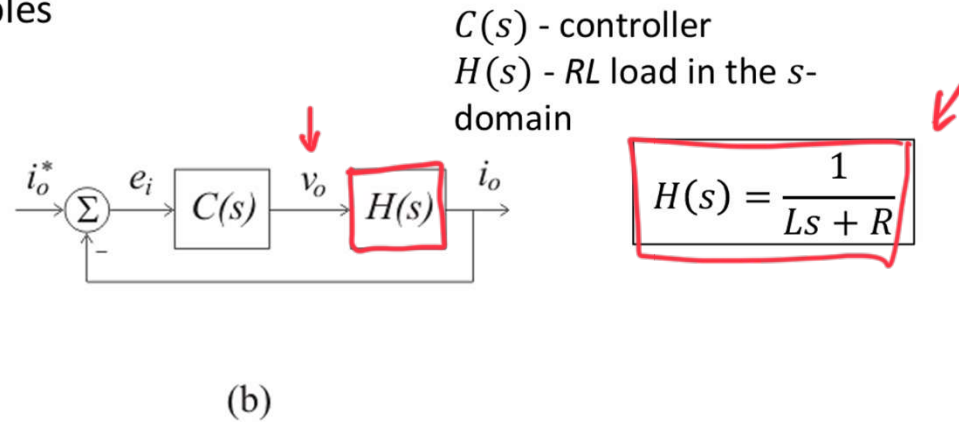
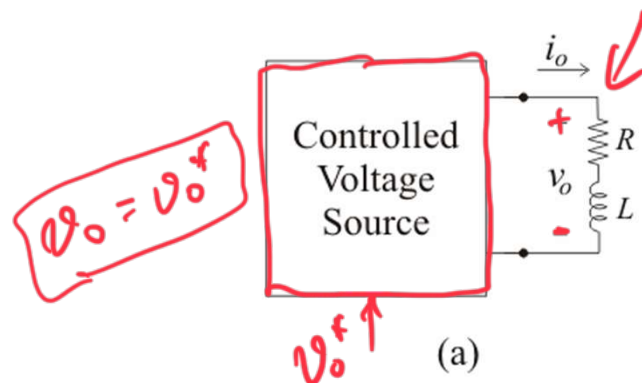
MATLAB codes



A dynamic simulation is an important aspect to test and validate control strategies

Control Strategies for Power Converters

2. Basic Control Principles



The output voltage (v_o) has been employed to reach this goal

A discretization method can be employed in order to simulate this system in a digital way

Euler discretization method

$$v_o = Ri_o + L \frac{di_o}{dt} \quad (1)$$

$$\frac{di_o}{dt} = \frac{v_o - Ri_o}{L} \quad (2)$$

$$i_o(t+h) = i_o(t) + \frac{v_o - Ri_o}{L} h \quad (3)$$



$$v_0 = R \cdot i_0 + L \frac{di_0}{dt}$$

$$L \frac{di_0}{dt} = v_0 - R \cdot i_0$$

$$\frac{\Delta i_0}{\Delta t} = \frac{1}{L} [v_0 - R \cdot i_0]$$

$$\Delta i_0 = \frac{1}{L} \cdot (v_0 - R \cdot i_0) \cdot h$$

$$i_0 = i_0 + \frac{1}{L} \cdot (v_0 - R \cdot i_0) \cdot h$$

while $t < t_{max}$,

$t = t + h_i$ ← STEP SIZE

$\vdots$ $\Delta t = h$

END $\Delta i_0 = \underline{i_{0f}} - \underline{i_{0i}}$

Control Strategies for Power Converters

2. Basic Control Principles

Implementation in open-loop operation

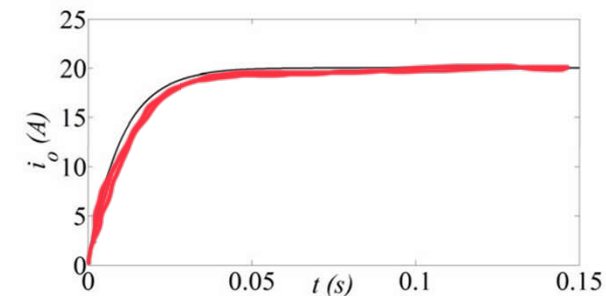
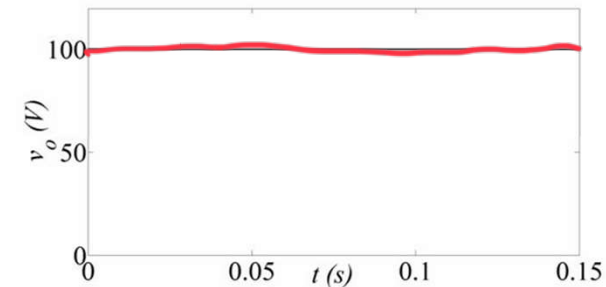


%Simulation of a RL load connected to a Controlled-Voltage-Source

R=5; %Resistance of the RL load
L=0.05; %Inductance of the RL load
io=0; %state variable (load current)
vo=100; %input variable (load voltage)
h=1e-4; %step size
j=0; %variable used to storage the outcomes
t=0; %initial value of the digital time
tmax=1; %maximum simulation time

while t<=tmax, %begining of the simulation loop
t = t + h; %time incrementation
io = io + (vo-R*io)*h/L; %state space variable 1
% storage of the variables
j = j + 1;
time(j)=t;
current(j)=io;
voltage(j)=vo;
end

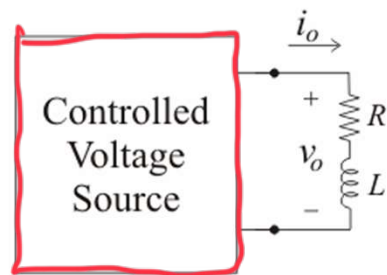
figure(1),plot(time,voltage)
figure(2),plot(time,current)



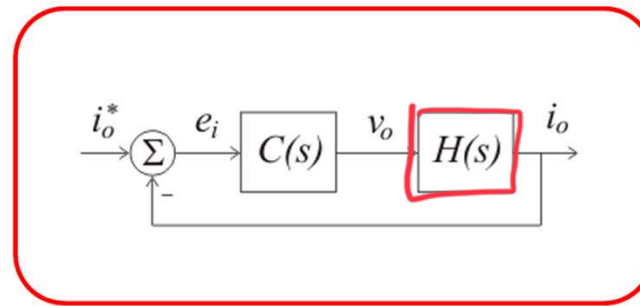
Control Strategies for Power Converters

2. Basic Control Principles

Closed loop operation



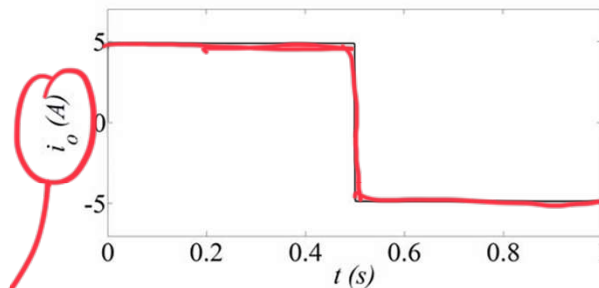
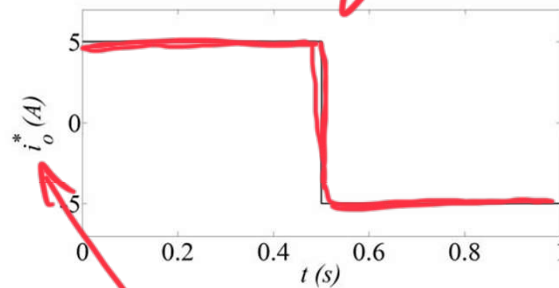
(a)



(b)

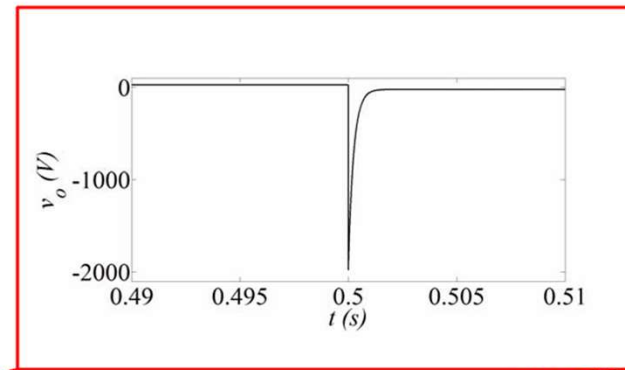
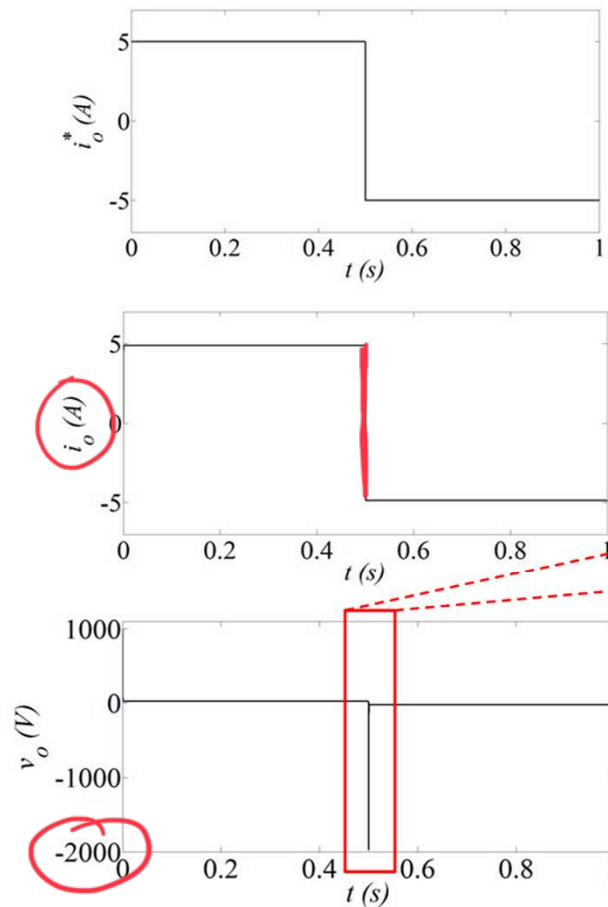
Proportional controller

$$v_o = K e_i \quad \Rightarrow \quad K = 200$$



Control Strategies for Power Converters

2. Basic Control Principles



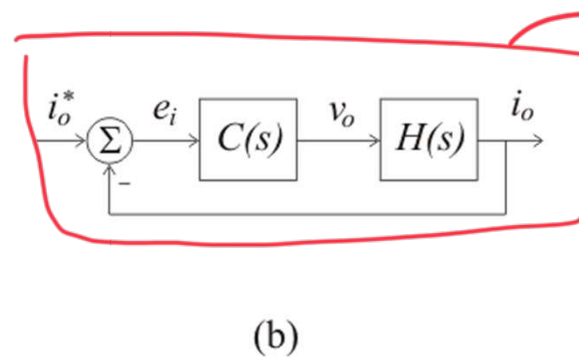
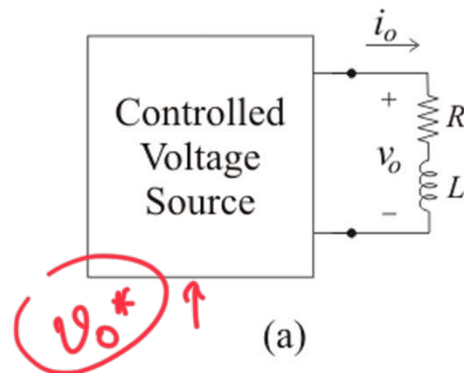
the control scheme presented in Fig. 9.1(b) is not showing the controlled voltage source (actuator) needed to implement this control scheme in a real world scenario

$$\frac{d}{dt}(L) = -\infty$$

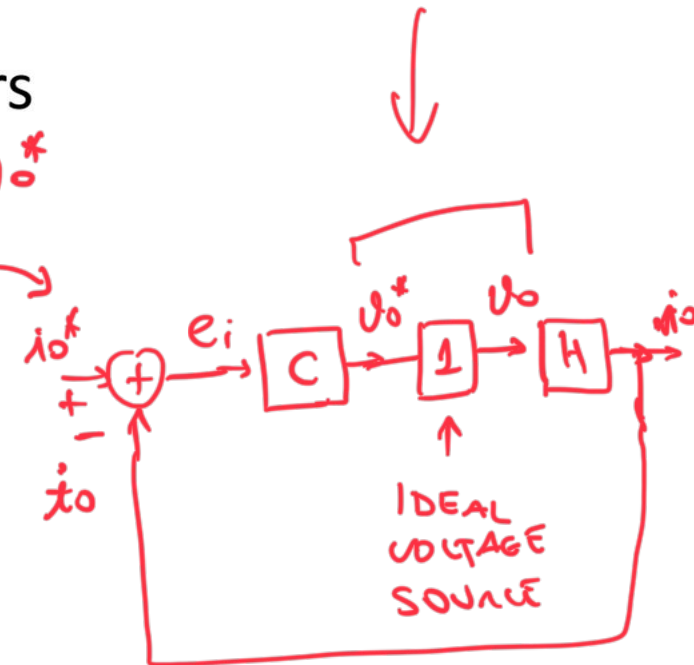
$$v_L = L \frac{di_o}{dt}$$

Control Strategies for Power Converters

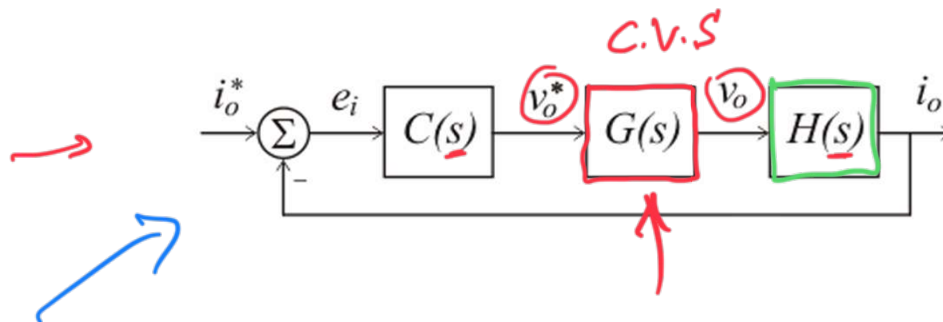
2. Basic Control Principles



$$v_o = v_o^*$$



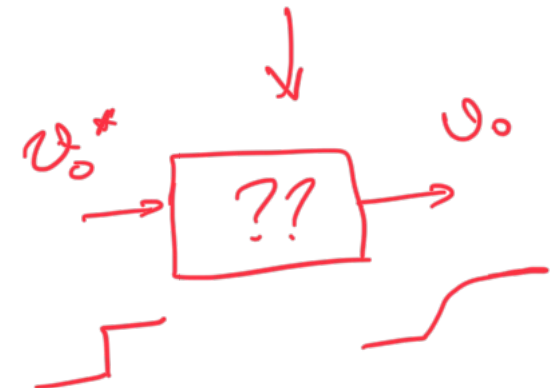
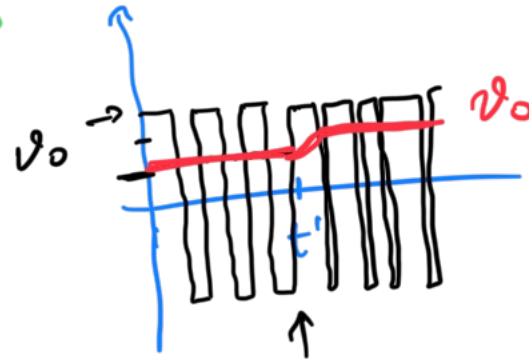
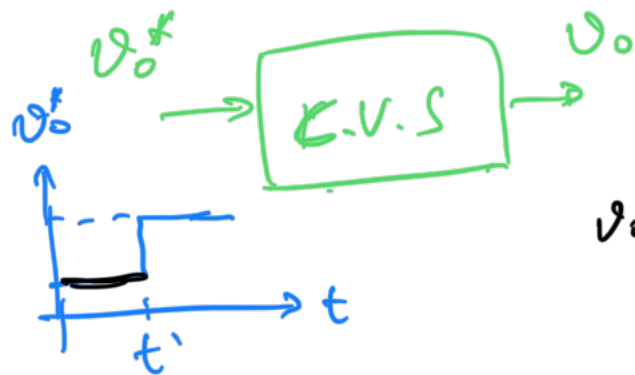
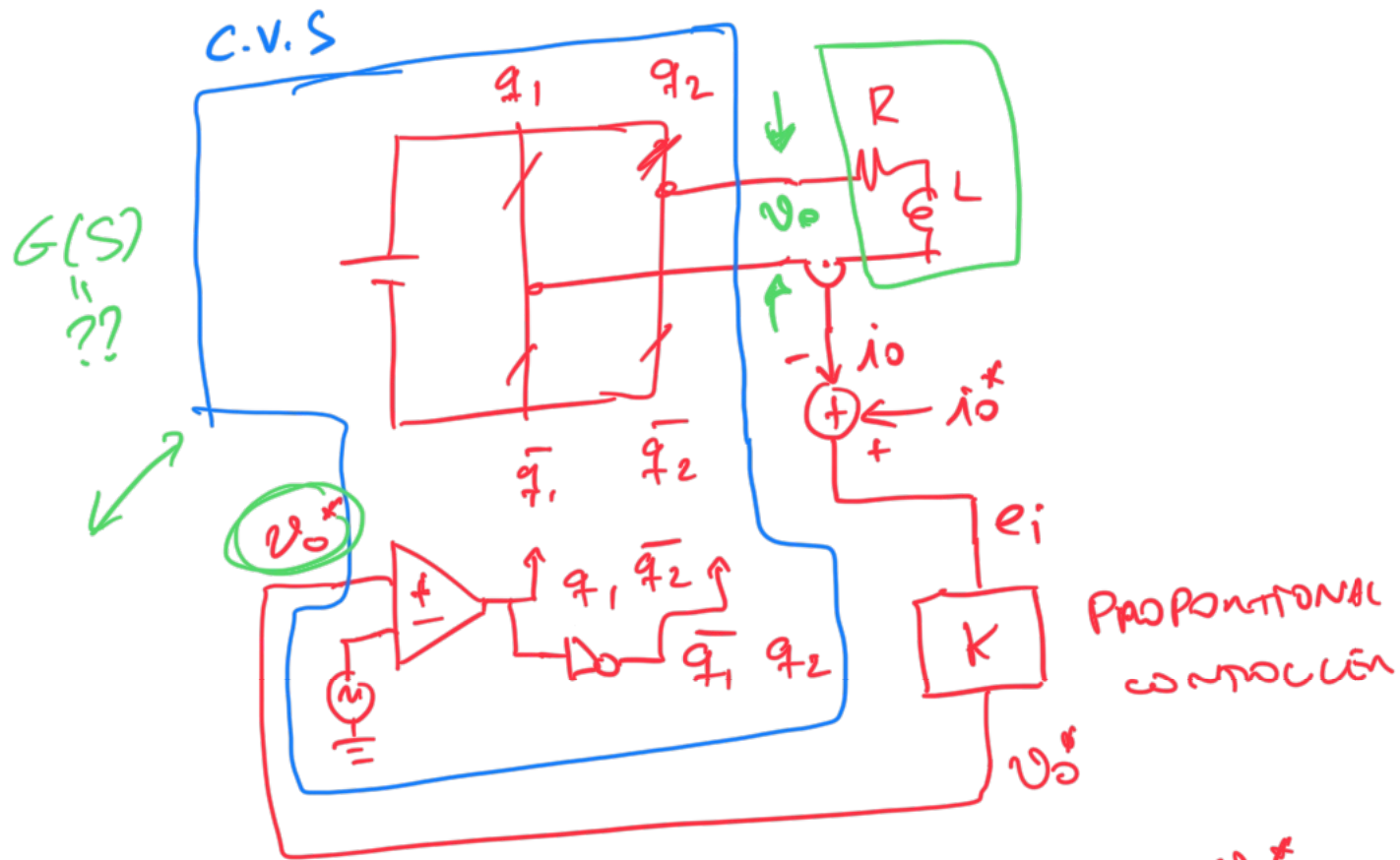
The control scheme presented in Fig. 9.1(b) is not showing the controlled voltage source (actuator) needed to implement the control scheme in a real world scenario.



$G(s) \longrightarrow$ H-bridge converter

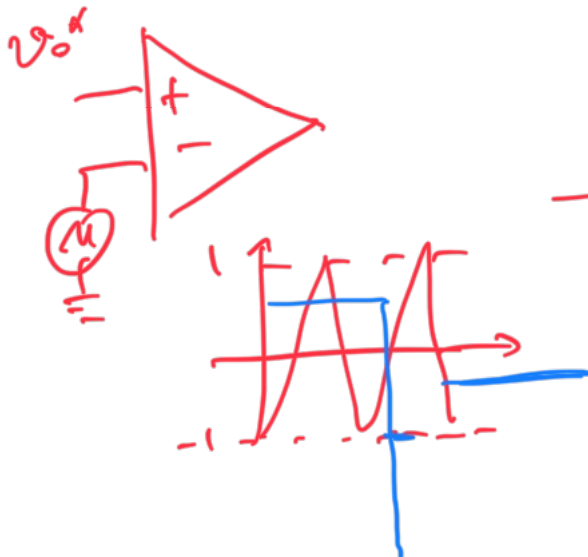
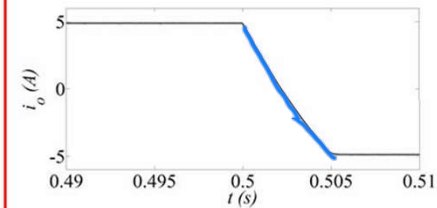
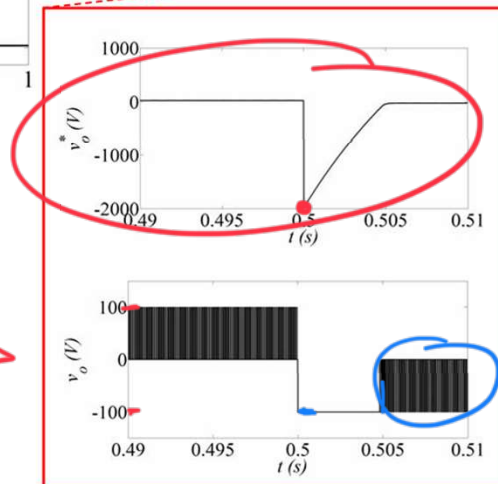
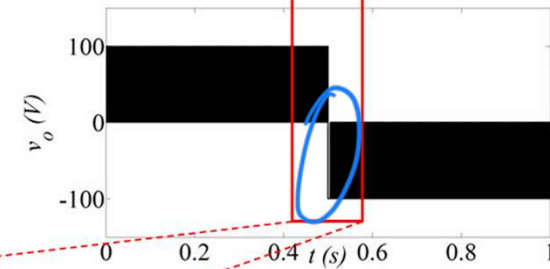
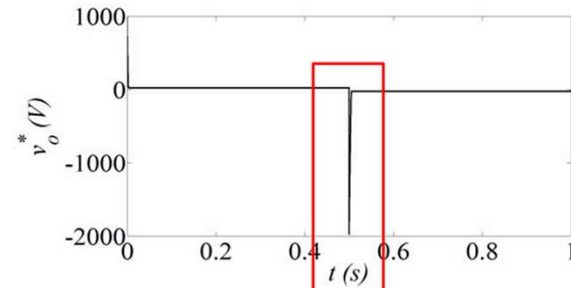
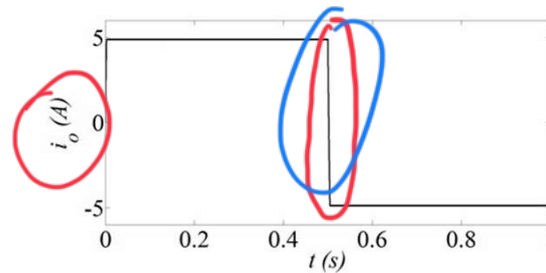
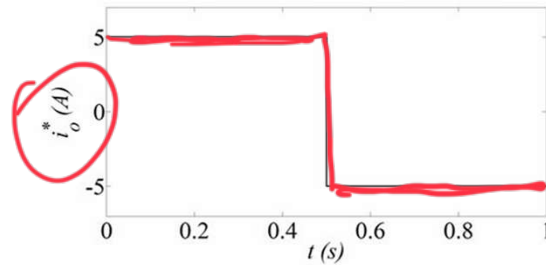
Ideal controlled voltage source
 $v_o = v_o^*$ with $G(s) = 1$

$$G(s) = \frac{1}{T_s s + 1} \quad T_s = 1/f_s$$



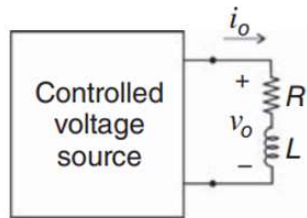
Control Strategies for Power Converters

2. Basic Control Principles



Problem 1

The figure below shows a controlled voltage source supplying voltage to an RL load. The MATLAB code used to simulate the system is presented as well.



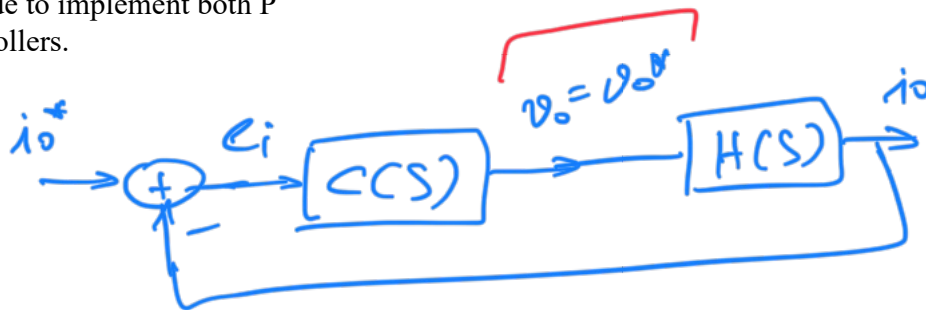
```
%Simulation of a RL load connected to a Controlled-
Voltage-Source
R=5; %Resistance of the RL load
L=0.05; %Inductance of the RL load
io=0; %state variable (load current)
vo=100; %input variable (load voltage)
h=1e-4; %step size
j=0; %variable used to storage the outcomes
t=0; %initial value of the digital time
tmax=1; %maximum simulation time

while t<=tmax, %begining of the simulation loop
    t = t + h; %time incrementation
    io = io + (vo-R*io)*h/L; %state space variable 1
    % storage of the variables
    j = j + 1;
    time(j)=t;
    current(j)=io;
    voltage(j)=vo;
end
figure(1),plot(time,voltage)
figure(2),plot(time,current)
```

$$X=0$$

$$\begin{aligned} e_i &= i_o^* - i_o; \\ X &= X + k_i \cdot e_i \cdot h; \\ v_o &= k_p \cdot e_i + X; \end{aligned}$$

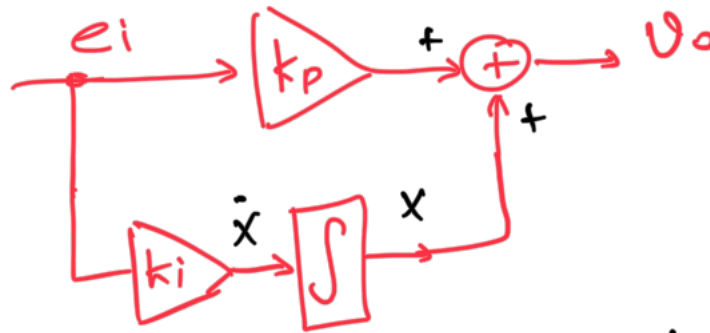
Draw a block diagram that represents a current controller implemented with an ideal CVS with: (a) proportional controller, and (b) PI controller. (c) Make changes to the provided code to implement both P and PI controllers.



$$\begin{aligned} C(s) &= K_P (P) \\ C(s) &= K_P + \frac{K_I}{s} (PI) \end{aligned}$$

$$\frac{V_o}{E_i} = k_p + k_i s^{-1}$$

$$V_o = k_p \cdot E_i + k_i \cdot s^{-1} \cdot E_i$$



$$\dot{x} = k_i \cdot e_i$$

$$\Delta t = h$$

$$\frac{dx}{dt} = k_i \cdot e_i$$

$$\frac{\Delta x}{\Delta t} = k_i \cdot e_i$$

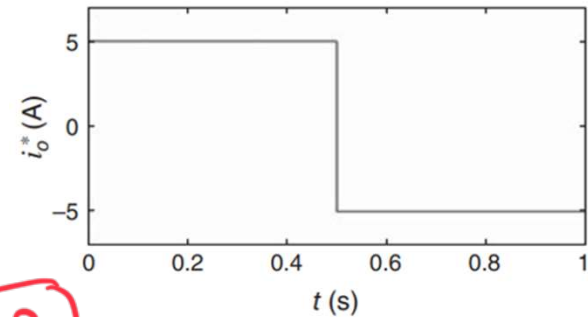
$$\Delta x = k_i \cdot e_i \cdot h$$

$$x = x + k_i \cdot e_i \cdot h$$

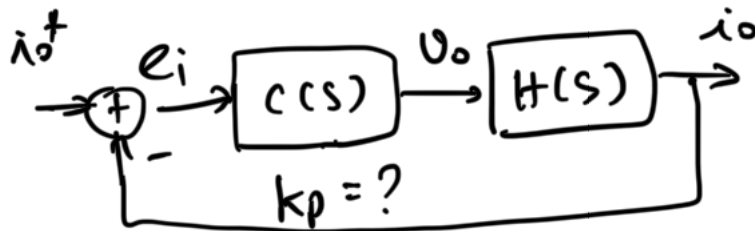
$$\begin{aligned} x &= x + k_i \cdot e_i \cdot h \\ \underline{u_o} &= k_p \cdot e_i + x \end{aligned} \quad (*)$$

Problem 2

- (a) For Problem 1 with a proportional controller, determine the gain K_p that will lead to an error of steady state condition of 0.025 if the reference current is 5 A.
- (b) For the gain obtained in (a), determine the maximum and minimum voltage v_o applied to the load if the reference current is shown aside. (c) Repeat (b) if a half-bridge converter is used with dc-link voltage of 500 V.



$K_p = ?$



$$i_o^* = 5 \text{ A}$$

$$e_i = 0.025 \text{ A}$$

$$v_o = K_p \cdot e_i \quad (1)$$

$$v_o = K_p \cdot (i_o^* - i_o)$$

$$K_p = \frac{v_o}{e_i} \quad (2)$$

$$\text{if } i_o = 0$$

$$v_o = K_p \cdot i_o^* \quad (3)$$

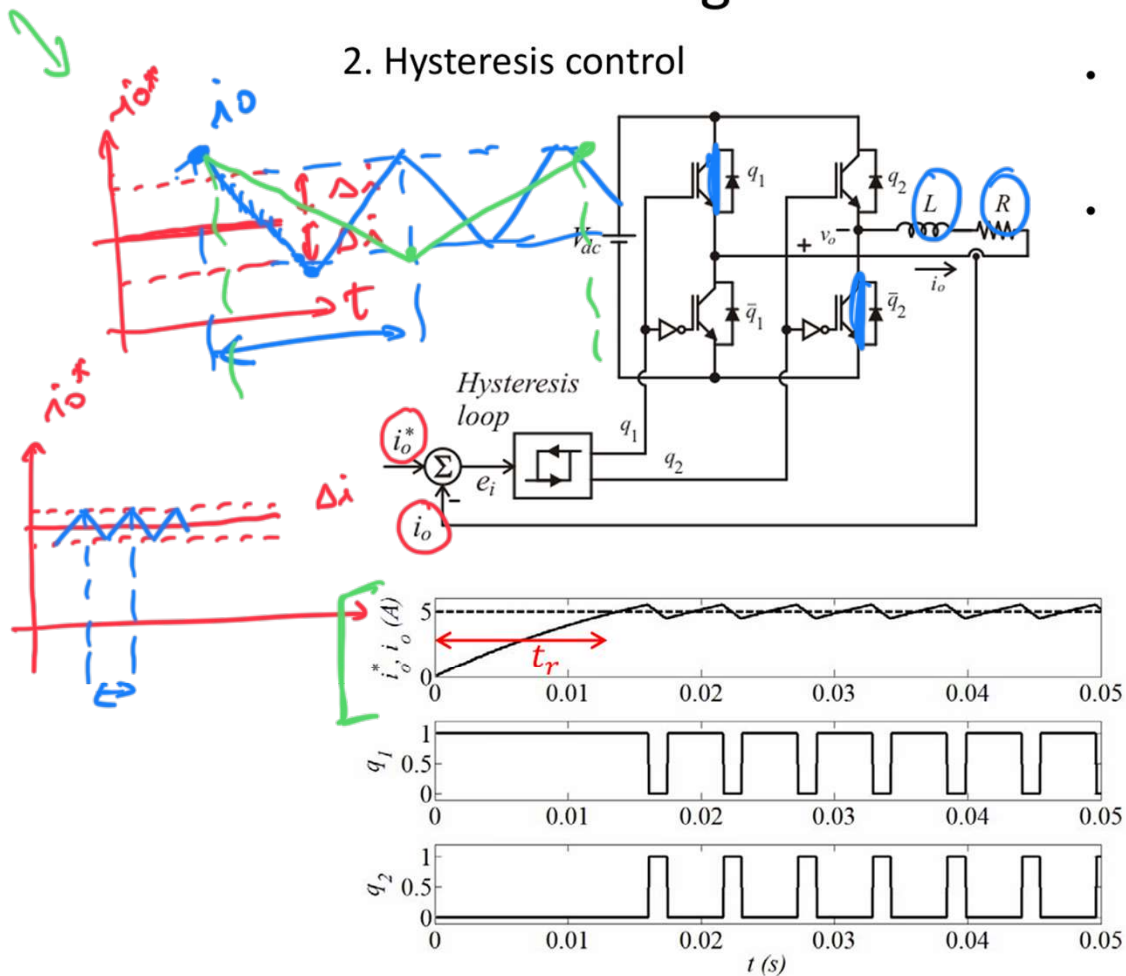
$$K_p = ?$$

$$e_i = 0$$

$$K_p \rightarrow \infty$$

Control Strategies for Power Converters

2. Hysteresis control



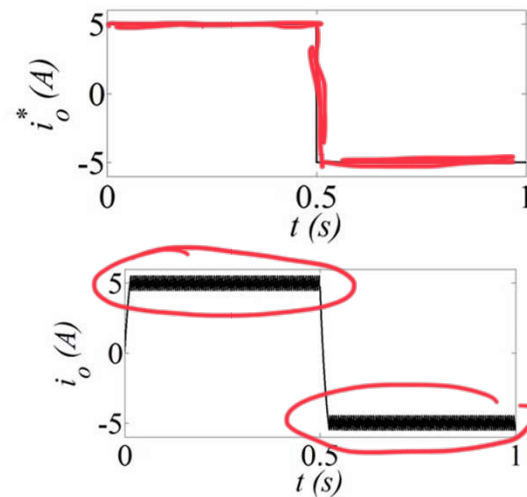
- If $i_o > i_o^* + \Delta i$, the load current must be reduced, hence

$$q_2 = on \text{ and } q_1 = off$$

- If $i_o < i_o^* - \Delta i$, the load current must be increased, hence

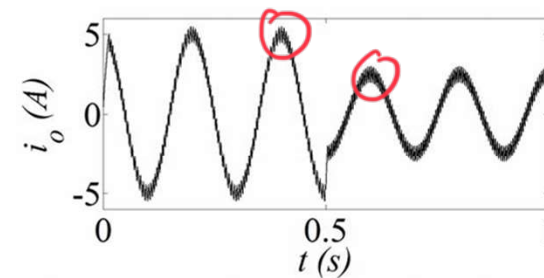
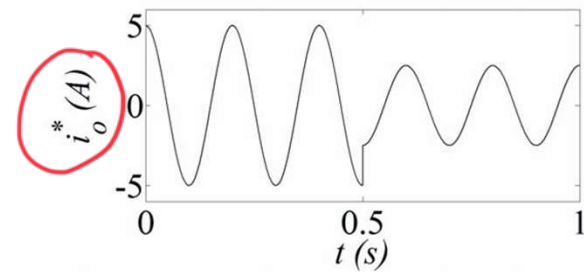
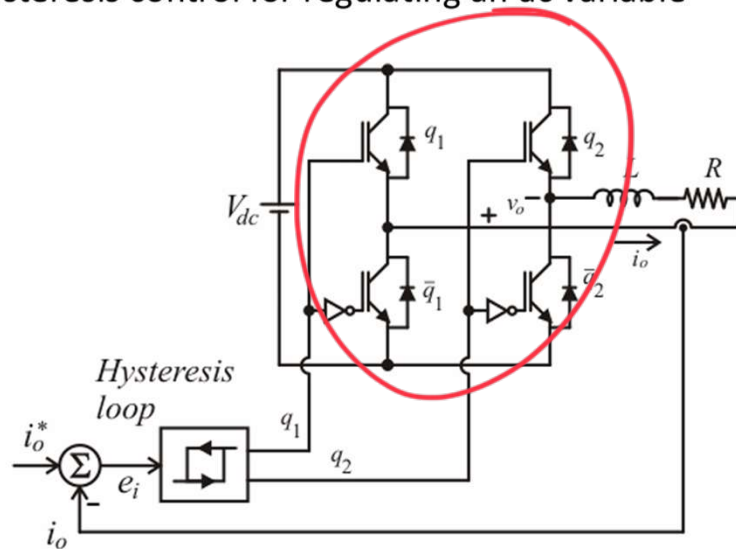
$$q_1 = on \text{ and } q_2 = off$$

$$t_r = \frac{L}{R} \ln\left(\frac{Ri_o^*}{V_{dc}} - 1\right)$$



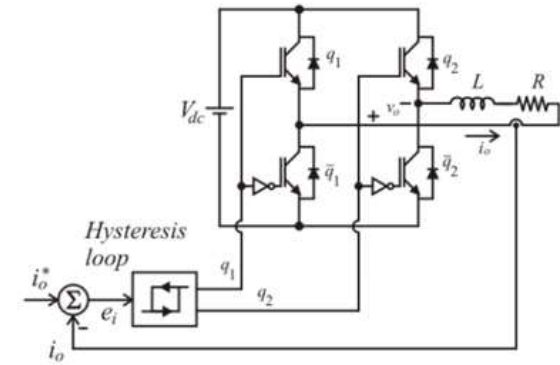
Control Strategies for Power Converters

Hysteresis control for regulating an ac variable



Problem 3

The power electronic converter under investigation exhibits a persistent short-circuit fault within its switch $\bar{q}_2$, characterized by a continuous conductive state irrespective of the applied gate control signal. (a) Change the hysteresis control logic of the other switches to guarantee that the measured current follows a constant reference current equal to 10 A with ripple of 2 A. Consider a very inductive load with inductance of 1 mH. (b) What is the minimum dc-link voltage to accomplish this control action. Assume a switching frequency equal to 10 kHz.



Control Strategies for Power Converters

2. Application of the hysteresis control for dc motor drive

$$v_f = R_f i_f + L_f \frac{di_f}{dt}$$

$$v_a = R_a i_a + L_a \frac{di_a}{dt} + e_a$$

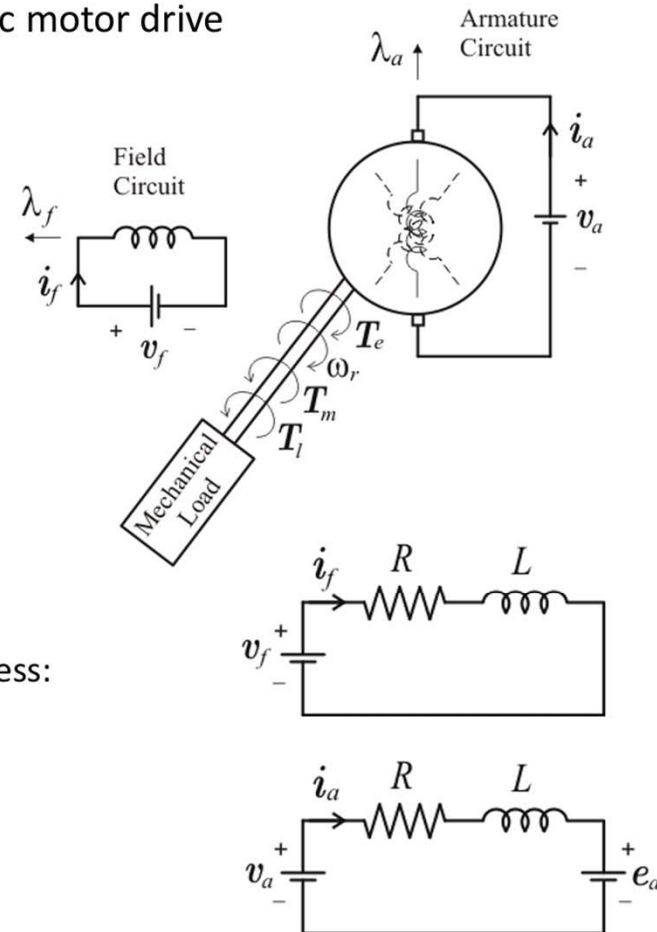
where $e_a = k_c \lambda_f \omega$ is the back *emf* (electromotive force); k_c is the coupling constant, λ_f is the field flux and ω is the mechanical speed

$$T_e = k_c \lambda_f i_a$$

Assuming no power loss in the conversion process:

$$T_e - T_l = B_m \omega + J_m \frac{d\omega}{dt}$$

where T_l is the load torque, B_m is the damping coefficient and J_m is the moment of inertia.

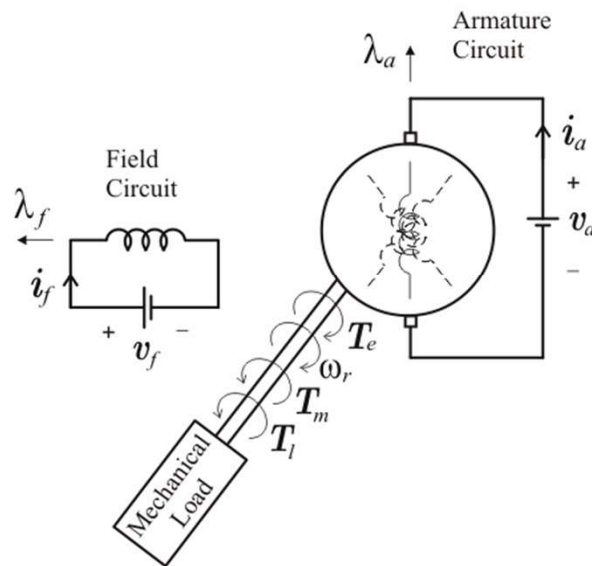


Control Strategies for Power Converters

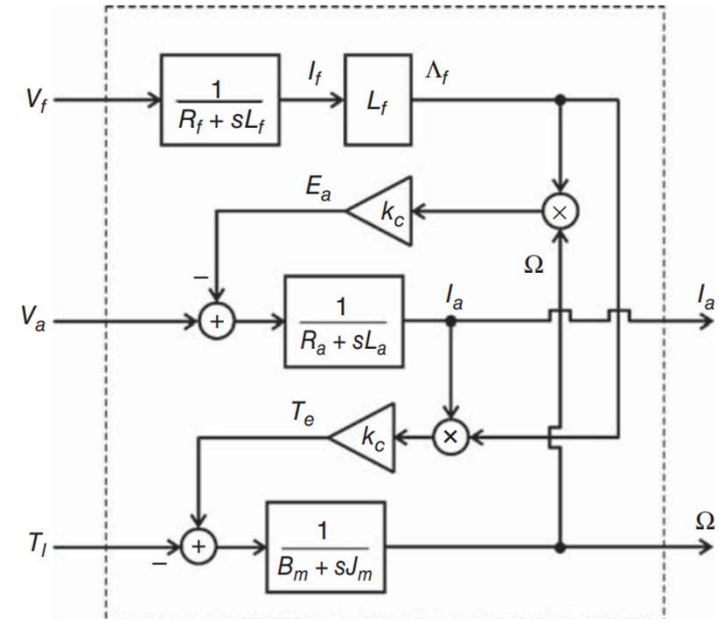
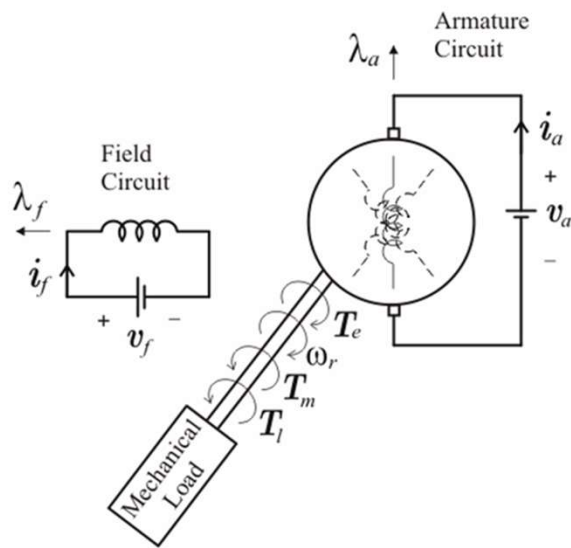
2. Application of the hysteresis control for dc motor drive

State space representation

$$\frac{d}{dt} \begin{bmatrix} i_a \\ \omega \end{bmatrix} = \begin{bmatrix} -R_a/L_a & -k_c \lambda_f/L_a \\ k_c \lambda_f/J_m & -B_m/J_m \end{bmatrix} \begin{bmatrix} i_a \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L_a & 0 \\ 0 & -1/J_m \end{bmatrix} \begin{bmatrix} v_a \\ T_l \end{bmatrix}$$



$$\frac{d}{dt} \begin{bmatrix} i_a \\ \omega \end{bmatrix} = \begin{bmatrix} -R_a/L_a & -k_c \lambda_f/L_a \\ k_c \lambda_f/J_m & -B_m/J_m \end{bmatrix} \begin{bmatrix} i_a \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L_a & 0 \\ 0 & -1/J_m \end{bmatrix} \begin{bmatrix} v_a \\ T_l \end{bmatrix}$$

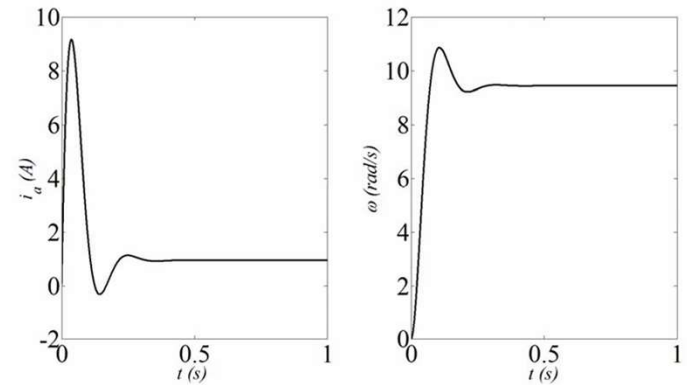


Control Strategies for Power Converters

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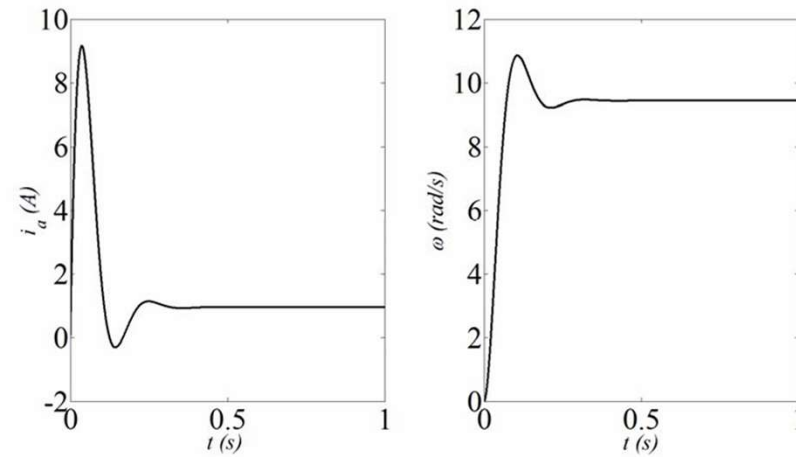
%constants
ra=0.06;
Laa=0.0018;
kf=0.1;
Bm=0.01;
Jm=0.001;
lambdaf=1;
%state variables
wr=0;
ia=0;
%input variables
va=1; %Armature voltage
Tl=.1; %Load torque
h=1e-4;%step size
j=0;
t=0;
tmax=2;
while t<=tmax,
t = t + h; % simulation time
ia=ia+(va/Laa-ra*ia/Laa-kf*lambdaf*wr/Laa)*h;%state
space 1
wr = wr + (kf*lambdaf*ia/Jm-Tl/Jm-Bm*wr/Jm)*h; %state
space 2
% storage the variables
j = j + 1;
time(j)=t;
armature_current(j)=ia;
rotor_speed(j)=wr;
end
figure(1),subplot(2,1,1),plot(time,armature_current)
subplot(2,1,2),plot(time,rotor_speed)
    
```

State space representation

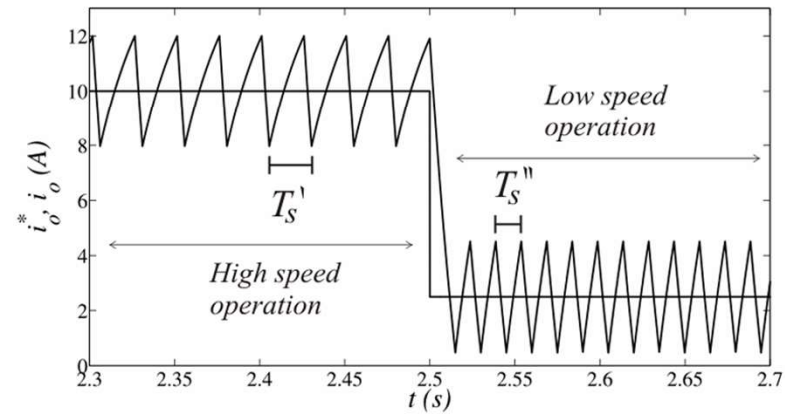
$$\frac{d}{dt} \begin{bmatrix} i_a \\ \omega \end{bmatrix} = \begin{bmatrix} -R_a/L_a & -k_c \lambda_f/L_a \\ k_c \lambda_f/J_m & -B_m/J_m \end{bmatrix} \begin{bmatrix} i_a \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L_a & 0 \\ 0 & -1/J_m \end{bmatrix} \begin{bmatrix} v_a \\ T_l \end{bmatrix}$$


Control Strategies for Power Converters

Open-loop operation:

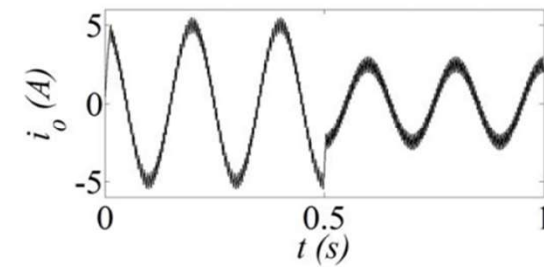
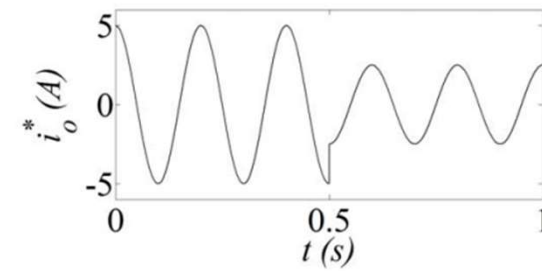
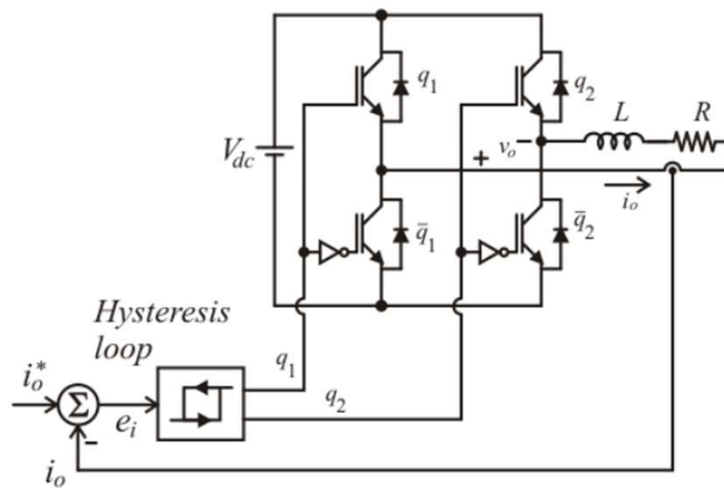


Close-loop operation:



Control Strategies for Power Converters

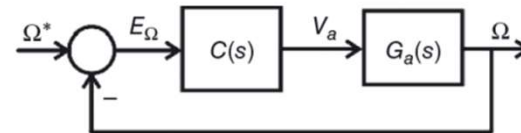
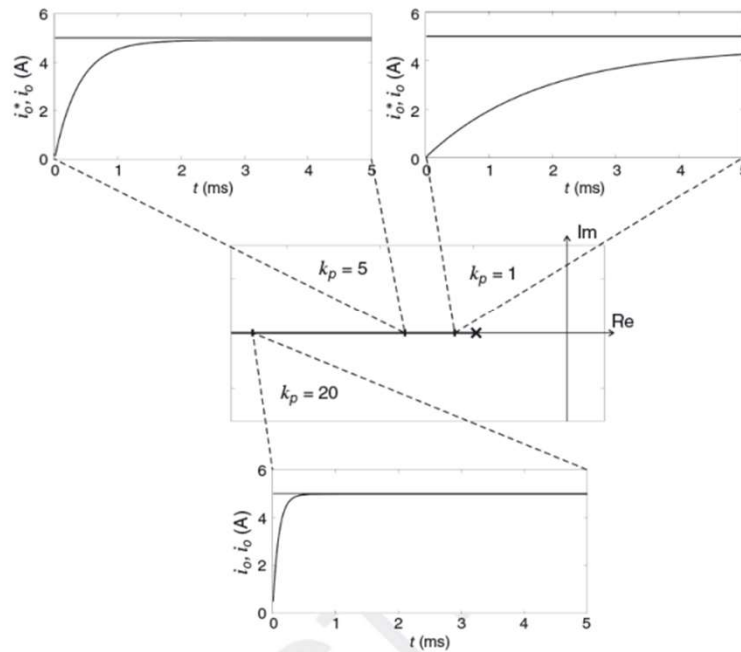
Hysteresis control for regulating an ac variable



Proportional Controller

$$F(s) = \frac{k_p}{R + sL}$$

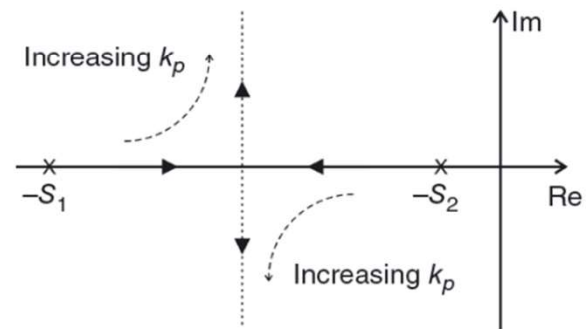
$$M(s) = \frac{k_p}{(R + k_p) + sL}$$



$$F(s) = \frac{k_p k_a}{(T_1 s + 1)(T_2 s + 1)}$$

$$M(s) = \frac{k_p k_a}{(T_1 s + 1)(T_2 s + 1) + k_p k_a}$$

Root locus of the dc motor drive system with proportional controller.

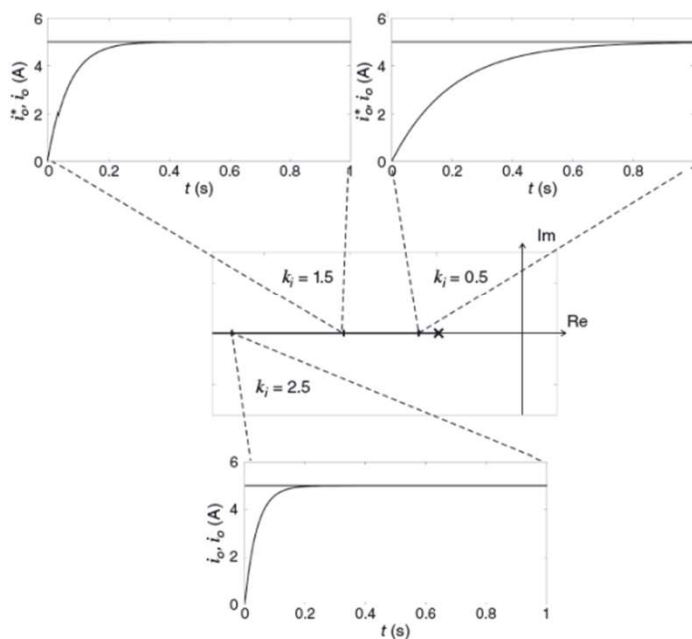


Proportional-Integral Controller

$$F(s) = \left(k_p + \frac{k_i}{s}\right) \left(\frac{1}{R + sL}\right)$$

$$F(s) = \frac{k_i}{s} \left(\frac{k_p}{k_i}s + 1\right) \left(\frac{1/R}{\frac{L}{R}s + 1}\right) \Rightarrow \frac{k_i}{k_p} = \frac{R}{L}$$

$$F(s) = \frac{k_i}{sR} \quad M(s) = \frac{k_i}{sR + k_i}$$

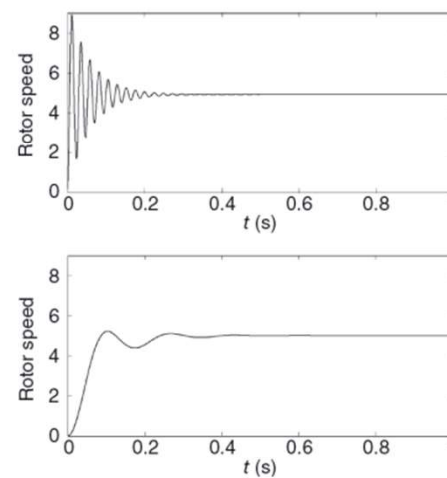


$$G_a(s) = \frac{k_a}{(T_1s + 1)(T_2s + 1)}$$

$$k_p/k_i = T_2$$

$$F(s) = \frac{k_i k_a}{s(T_1s + 1)}$$

$$M(s) = \frac{k_i k_a}{s(T_1s + 1) + k_i k_a}$$



Proportional-Integral-Derivative Controller

$$C(s) = k_p + \frac{k_i}{s} + k_d s$$

$$C(s) = k_p + \frac{k_i}{s} + \frac{k_d s}{sT_d + 1}$$

$$C(s) = \frac{(T_d k_p + k_d)s^2 + (T_d k_i + k_p)s + k_i}{s(T_d s + 1)}$$

$$k_i = \frac{1}{T_i} = \frac{1/k_a}{4T_d}$$

$$k_p = \frac{T_1 + T_2 - T_d}{T_i}$$

$$k_d = \frac{T_1 T_2 - (T_1 + T_2 - T_d)T_d}{T_i}$$

The parameters k_p and k_d can be employed to allocate the closed loop poles anywhere in the complex plane.

LINEAR CONTROL – ac VARIABLE

Resonant PI type of controller

$$C(s) = \frac{k_a s^2 + k_b s + k_c}{(s^2 + \omega^2)}$$

state space representation of this controller

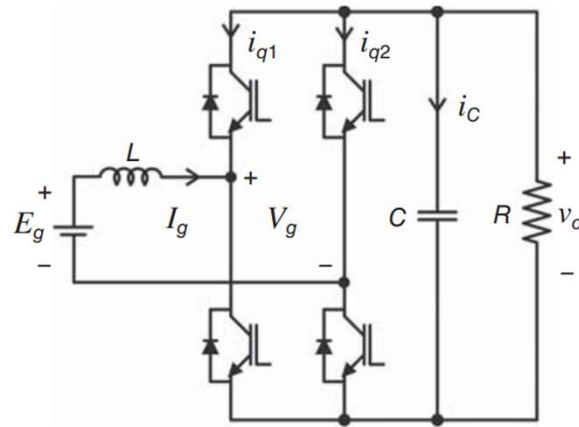
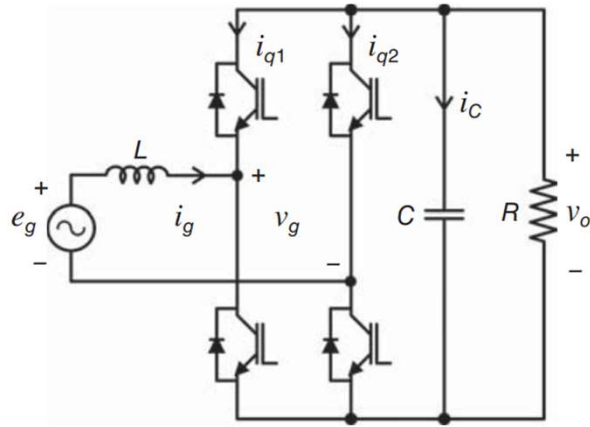
$$\dot{x}_1 = x_2 + k_b e$$

$$\dot{x}_2 = -\omega^2 x_1 + k_d e$$

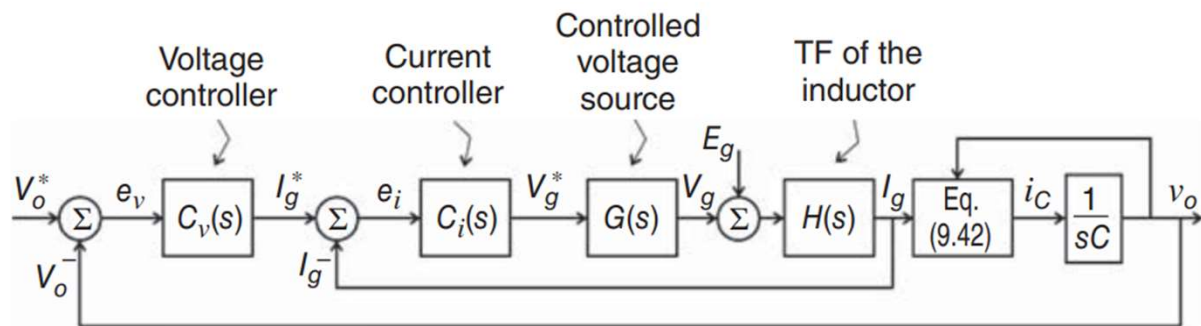
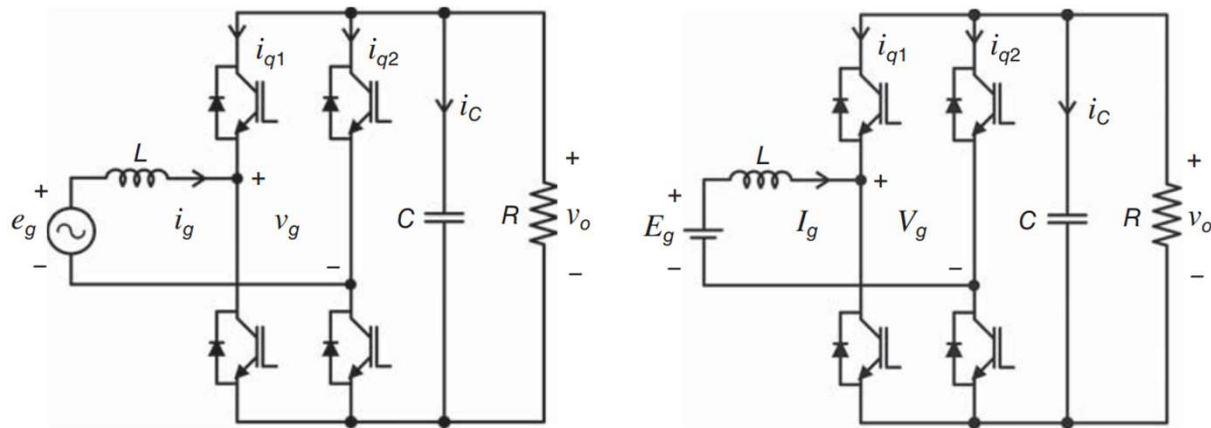
$$y = x_1 + k_a e$$

$$k_d = k_c - \omega^2 k_a$$

CASCADE CONTROL STRATEGIES



CASCADE CONTROL STRATEGIES



CASCADE CONTROL STRATEGIES

